

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY

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Course: 21CSC203P — Advanced Programming Practice	Department: Computer Science and Engineering

Unit I | Session 1 | SLO 1: Elements of Programming Languages (Code: 1011)

Q: 1. What is the syntax and semantics of the following Java statement: `int x = 5 + 3;`?

Ans: Syntax: The statement consists of a type specifier (`int`), an identifier variable name (`x`), the assignment operator (`=`), an arithmetic expression (`5 + 3`), and a statement terminator semicolon (`;`). It complies strictly with Java's Context-Free Grammar (CFG) rules for local variable declaration and initialization.

Semantics: At runtime, memory is allocated on the stack for a 32-bit signed two's complement integer. The binary addition operator computes the arithmetic sum of operands 5 and 3, yielding 8. The value 8 is then stored in the memory location referenced by variable `x`.

Q: 2. Identify lexical tokens in a simple Java program.

Ans: Consider the statement: `int count = 10;`

- **Keyword:** `int` (reserved word defining data type)
- **Identifier:** `count` (user-defined variable name)
- **Operator:** `=` (assignment operator)
- **Literal:** `10` (integer constant literal)
- **Separator/Punctuator:** `;` (statement terminator delimiter)

Q: 3. Modify a sample program to demonstrate the use of grammar rules in Java.

Ans: Java Demonstration:

Grammar rules dictate that statements must be inside methods, variables must be typed, and block scopes must be balanced.

```
public class GrammarDemo {  
    public static void main(String[] args) {  
        // Rule 1: Type declaration preceding identifier  
        int base = 10;  
        int height = 5;  
        // Rule 2: Valid arithmetic expression and semicolon termination  
        double area = 0.5 * base * height;  
        // Rule 3: Balanced curly braces for block scope  
        if (area > 0) {  
            System.out.println("Calculated Area: " + area);  
        }  
    }  
}
```